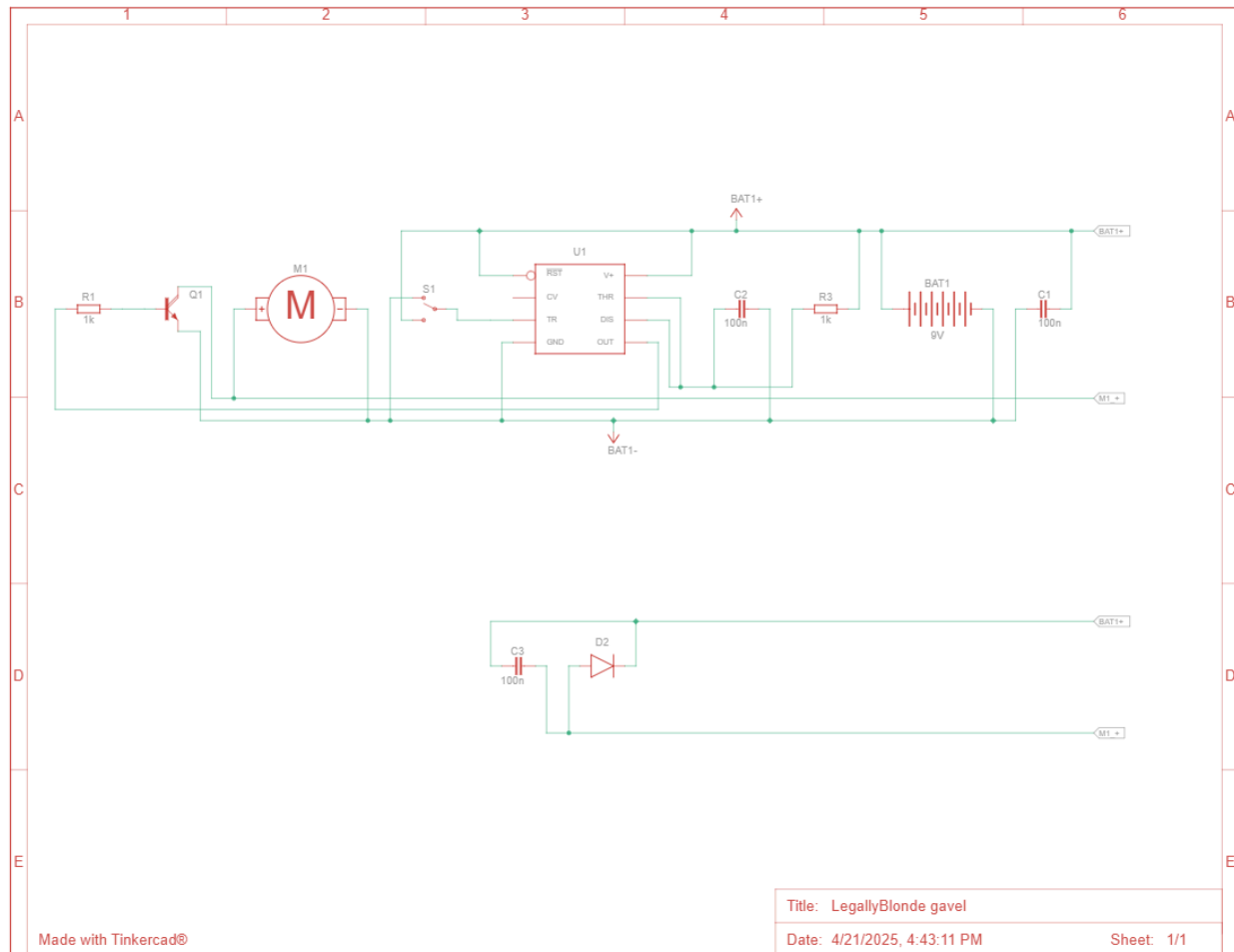
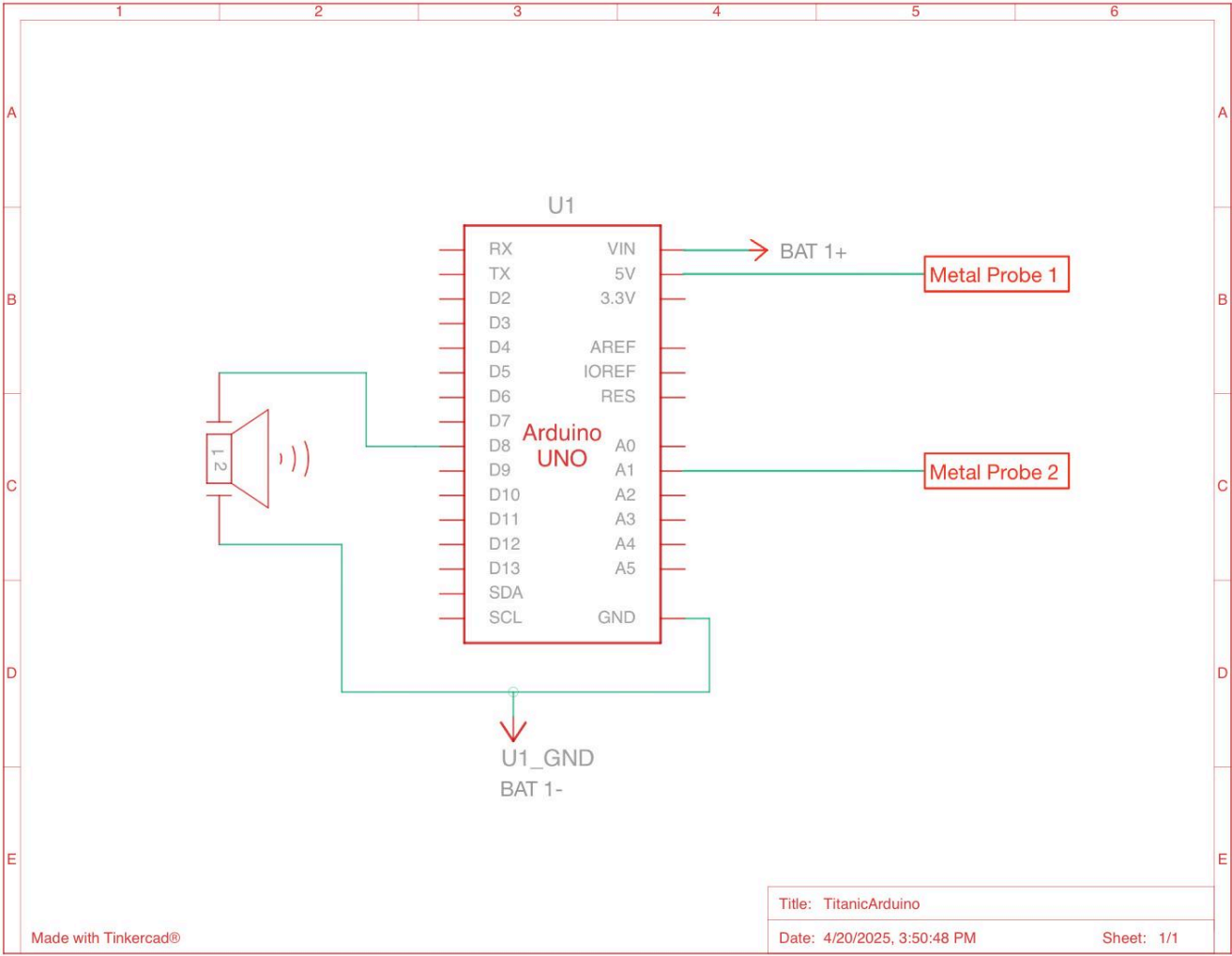


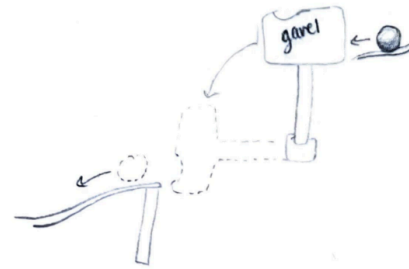
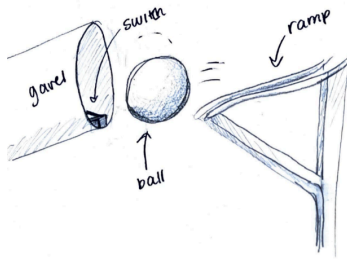
Nora Atkins, Sienna Giuseppi, Sara Meran, and Abby Mirsky



Titanic Arduino Circuit Schematic:



SportBot Action Summary:



Input Action/Trigger

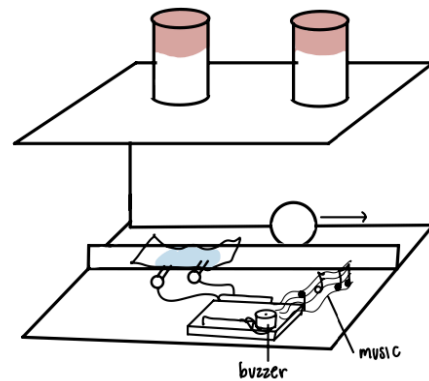
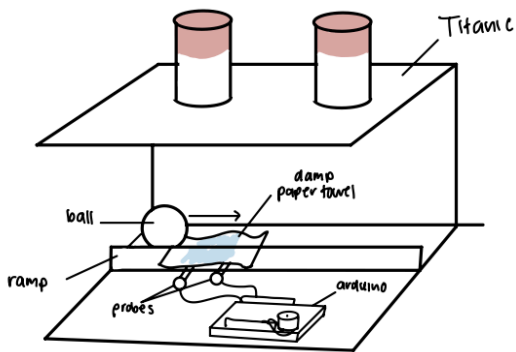
Circuit Function

Output Action

1 Ball rolls from ramp into a tunnel (the gavel) and hits a switch

1 Circuit is activated by the switch and uses a 555 timer which causes motor to rotate for 2 seconds

1 Motor, which is connected to the gavel, spins to rotate gavel and transport the ball onto another ramp



2 Ball pushes wet paper towel down onto 2 metal probes

2 circuit connects and upon a high liquid sensor reading, buzzer turns on

2 The Titanic song, "My Heart Will Go On" plays for 10 seconds

Team Reflection

Worked Well

During this project, the team was able to consistently communicate well through both a sub-team (4 person) and larger team (14 person) group chat. The sub-team group chat was used to schedule small group meetings on Thursdays at 9 am at Marketplace, clarify design details of the two sport bots, and send updates on individual progress. The larger team group chat was used to arrange large-scale meetings to integrate the different sport bots. These meetings generally occurred in the morning on weekends when most group members were free. Additionally, the group's time management was a strong feature of this group, as both sport bots were completed with ample time before the due date. Despite initial inconsistencies, each sport bot had numerous successful test runs that ensured smooth execution on test day. This success can be attributed to the team's early start and steady work on their circuits!

Challenges Faced

One challenge faced in the group was that some of the group members had very different work styles and times. For example, some of the group members work best and are most productive late at night whereas other group members work best in the mornings or early evenings. To combat this, the group stuck to the initial plan that Abby and Sienna would work predominantly on the Legally Blonde portion of the sportbot and Sara and Nora would focus on the Titanic portion of the sportbot. This way, Abby and Sienna could work at night and Sara and Nora could work earlier in the day. In order to promote communication and foster collaboration among the four group members, everybody came together at the 9 AM meetings to regroup, understand the progress, and discuss what improvements are to be made moving forward. The group chat was also consistently in use to communicate what work had been done individually and what work is remaining.

Another challenge faced was inconsistencies with circuit functionality. After the circuits were designed and tested on TinkerCad, it was time to build them in person. For the Titanic circuit, one difficulty was putting together a water/liquid sensor since the Foundry and kits did not include any. This required the team to test how different types of nails and screws responded to liquid. After lots of testing and iterations of installing and reinstalling, the final design for the metal probes was settled on. For the Legally Blonde circuit, there was much trial and error involved with figuring out the timing of the 555 circuit to make the motor turn just enough to deliver the ball to the next rail without turning too far/quickly or stopping too short. Also, since this bot involved moving the ball between separate tracks using the gavel, designing the housing for such a circuit presented issues when navigating the wiring and other physical components. Ensuring the design could rotate the gavel without pulling or tangling wires in the circuit required many iterations. Additionally, 3D printed parts that held the motor in place became worn after countless trials and had to be replaced, which also contributed to further iterations of the design.

Finally, for both the Titanic and Legally Blonde circuits, there was a lot of testing and troubleshooting in order to make the SportBot as consistent as possible. There were cases where sometimes the circuit would function as intended and other times where it wouldn't. After much testing and refining of sensor values, both the Titanic and Legally Blonde aspects of the SportBot were working successfully and had repeatable, consistent outputs (the turning of the gavel for Legally Blonde and the music for the Titanic)

Lessons Learned

During this project, the team learned the importance of starting a project early. Having a strong initial start propelled this team to success, as there was ample time for testing and debugging of the circuit. Both of our team's bots were finished ahead of schedule and were ready to go before most of the other SportBots in the larger group. This gave the team the opportunity to help other groups and help add decorations to enhance the overall project, as well as providing us ample time to work out any issues with connecting tracks between other team's bots.

Additionally, the team learned valuable skills about how to divide work efficiently. Specifically, Abby and Sienna were able to focus on the Legally Blonde themed sport bot, whereas Nora and Sara tackled The Titanic themed sport bot. Having two people be responsible for each circuit drastically cut down the time it took to build both bots, and the sub teams within our team allowed for more flexibility in building times and styles.

Growth and Improvements

Over the course of the project, the team members improved their electronics and circuit skills. For some of the group members, it was their first time using electronics to accomplish a goal that they set. Having the independence to decide on the circuit input and output caused the group members to gain a lot of electronic skills and more confidence for handling circuits. This knowledge and confidence will be immensely helpful in the future as the group members continue to pursue various electronic based projects. In the future, the group would have more communication and coordination for the sizing and materials used to create the SportBot tracks. Additionally, though our team did communicate well within the larger group and between both sub-teams, something to think about for the future is more thought-out division of labor for our individual bot so all members can play more to their strengths and spend equal time working on the bot. While both the Titanic and Legally Blonde aspects of the SportBot were very decorated and a lot of care was placed in their design, the group could have done a better job at creating more of a unified and coherent style for the SportBot. Additionally, another improvement would have been simply having more communication and collaborative troubleshooting throughout the project; while this was not the case with our team, other teams did finish at the last minute which delayed the overall group's progress. Overall, this SportBot team worked very hard and worked well together in order to create a successful final product!